CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

IT342-Section
SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: User Register and Authentication

Prepared By: Theo Cedric Chan

Date of Submission: 2/6/2026

Version: 1.5

Table of Contents

1. Introduction....3
   1.1. Purpose....3
   1.2. Scope....3
   1.3. Definitions, Acronyms, and Abbreviations....3
2. Overall Description....3
   2.1. System Perspective....3
   2.2. User Classes and Characteristics....3
   2.3. Operating Environment....3
   2.4. Assumptions and Dependencies....3
3. System Features and Functional Requirements....3
   3.1. Feature 1:....3
   3.2. Feature 2:....3
4. Non-Functional Requirements....3
5. System Models (Diagrams)....4
   5.1. ERD....4
   5.2. Use Case Diagram....4
   5.3. Activity Diagram....4
   5.4. Class Diagram....4
   5.5. Sequence Diagram....4
6. Appendices....4

1. Introduction

1.1. Purpose

The purpose of this system is to provide a secure and centralized User Registration and Authentication platform for students and staff within a school environment. This document is intended for project stakeholders, developers, and system architects to ensure the technical implementation matches the project requirements.

1.2. Scope

The system will manage user identities by allowing guests to register and authenticated users to log in, view their profiles, and manage account settings like theme and password changes. Its boundaries are limited to account management and basic dashboard browsing within a school-specific domain.

1.3. Definitions, Acronyms, and Abbreviations

- **ERD:** Entity Relationship Diagram.
- **(PK) / (FK):** Primary Key / Foreign Key.
- **Authentication:** The process of verifying the identity of a user.
- **Authenticated User:** A user who has successfully logged into the system.

2. Overall Description

2.1. System Perspective

This system acts as the entry point for a larger educational platform. It relies on a relational database to link users to their respective schools via a school_id

2.2. User Classes and Characteristics

- **Guest (User):** Unauthenticated individuals who can only access the registration and login modules.
- **Authenticated (User):** Logged-in students or staff who can access the main dashboard, view/edit profiles, and modify settings.

2.3. Operating Environment

The system will be a cross-platform application designed to provide a seamless experience across both **Web and Mobile (Android)** platforms.

- **Web:** Accessible via modern web browsers (Chrome, Safari, Edge, Firefox) using a responsive UI.
- **Mobile:** A dedicated mobile application (or Progressive Web App) optimized for touch interfaces.
- **Backend:** A centralized server and API to synchronize user data across both web and mobile sessions.
- **Database:** A relational database (SQL-based) to store the `Users` and `School` entity relationships.

2.4. Assumptions and Dependencies

2.4.1 Assumptions

- **Connectivity:** It is assumed that users have a stable internet connection for real-time authentication and profile updates.
- **Device Compatibility:** It is assumed that users are using devices that support modern web standards or minimum OS versions (e.g., Android 10+ or iOS 15+).
- **Data Accuracy:** We assume the School table is pre-populated with valid email_domain data to facilitate registration filtering.

2.4.2 Dependencies

- **School Email Systems:** Registration functionality depends on the availability and validity of the school-issued email addresses for student verification.
- **Cloud Storage:** The storage of profile_picture files depends on an external cloud storage provider (like AWS S3 or Firebase Storage).
- **API Availability:** The mobile app depends on the consistent availability of the REST or GraphQL API to communicate with the central database.

3. System Features and Functional Requirements

Describe each major feature of the system and its functional requirements.

3.1. Feature 1: User Registration and Secure Login

- **Description:** Allows new users to create an account and existing users to access the dashboard.
- **Functional Requirements:**
  - The system shall validate the user's email against the school's authorized email_domain.
  - The system shall store a hashed version of the user's password for security.
  - The system shall provide a unique user_id for every successful registration.

3.2. Feature 2: Account and Settings Management

- **Description:** Enables authenticated users to manage their personal data and interface preferences.
- **Functional Requirements:**
  - The system shall allow users to view and update their profile picture and contact details.
  - The system shall provide an "extend" feature for users to change their account password securely.

- The system shall allow users to toggle interface themes (e.g., Light/Dark mode).

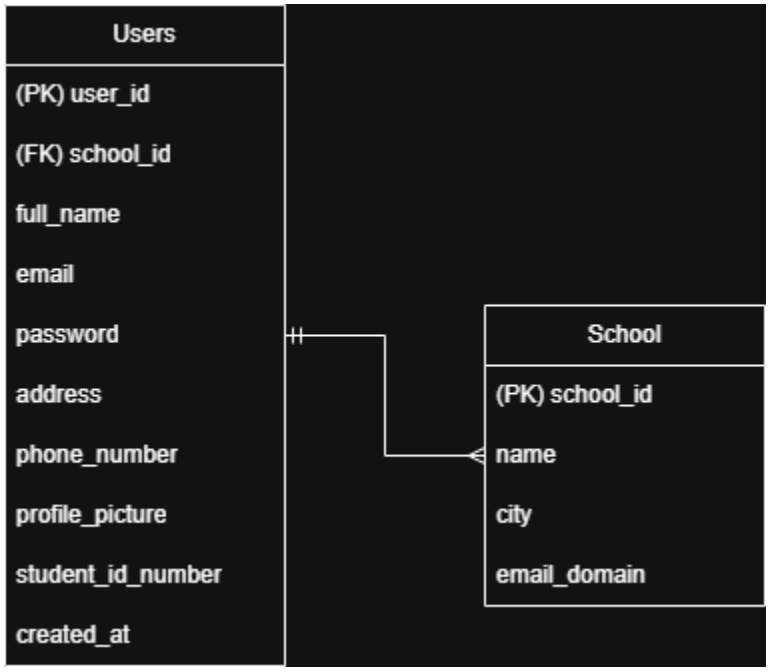
4. Non-Functional Requirements

- **Security:** Passwords must be encrypted, and sessions must be terminated upon a "Logout" action to prevent unauthorized access.
- **Usability:** The navigation flow from the dashboard to profile details must follow a logical path as defined in the activity diagram.
- **Reliability:** The system must maintain data integrity between the Users and School tables during registration.

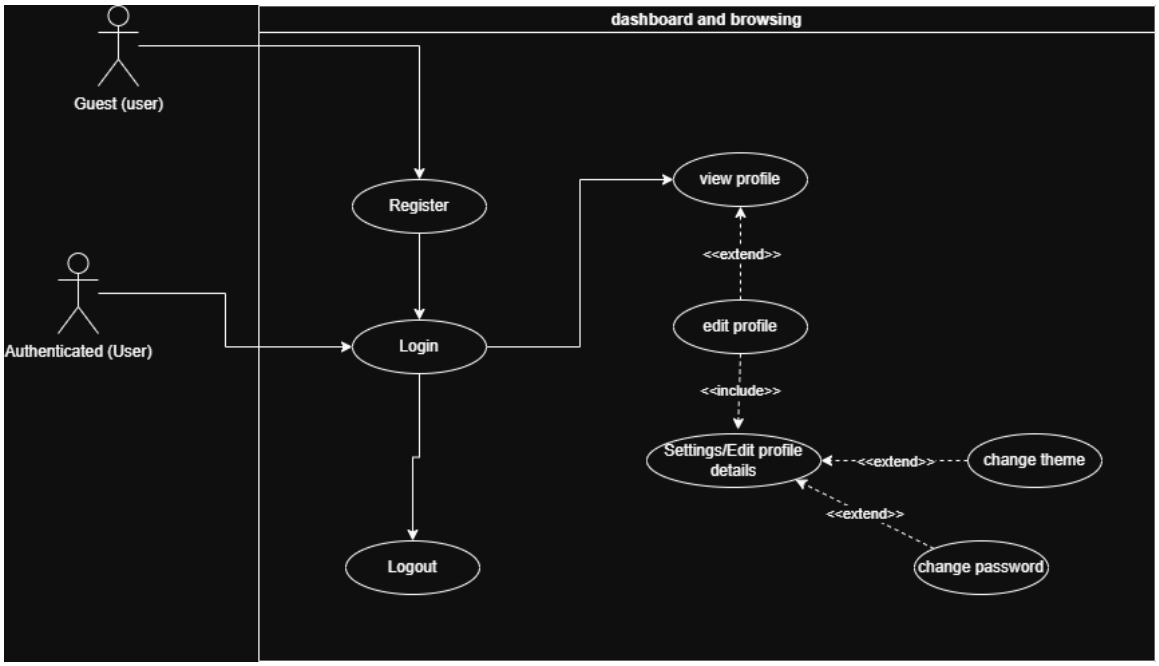
5. System Models (Diagrams)

Insert the necessary diagrams for the system:

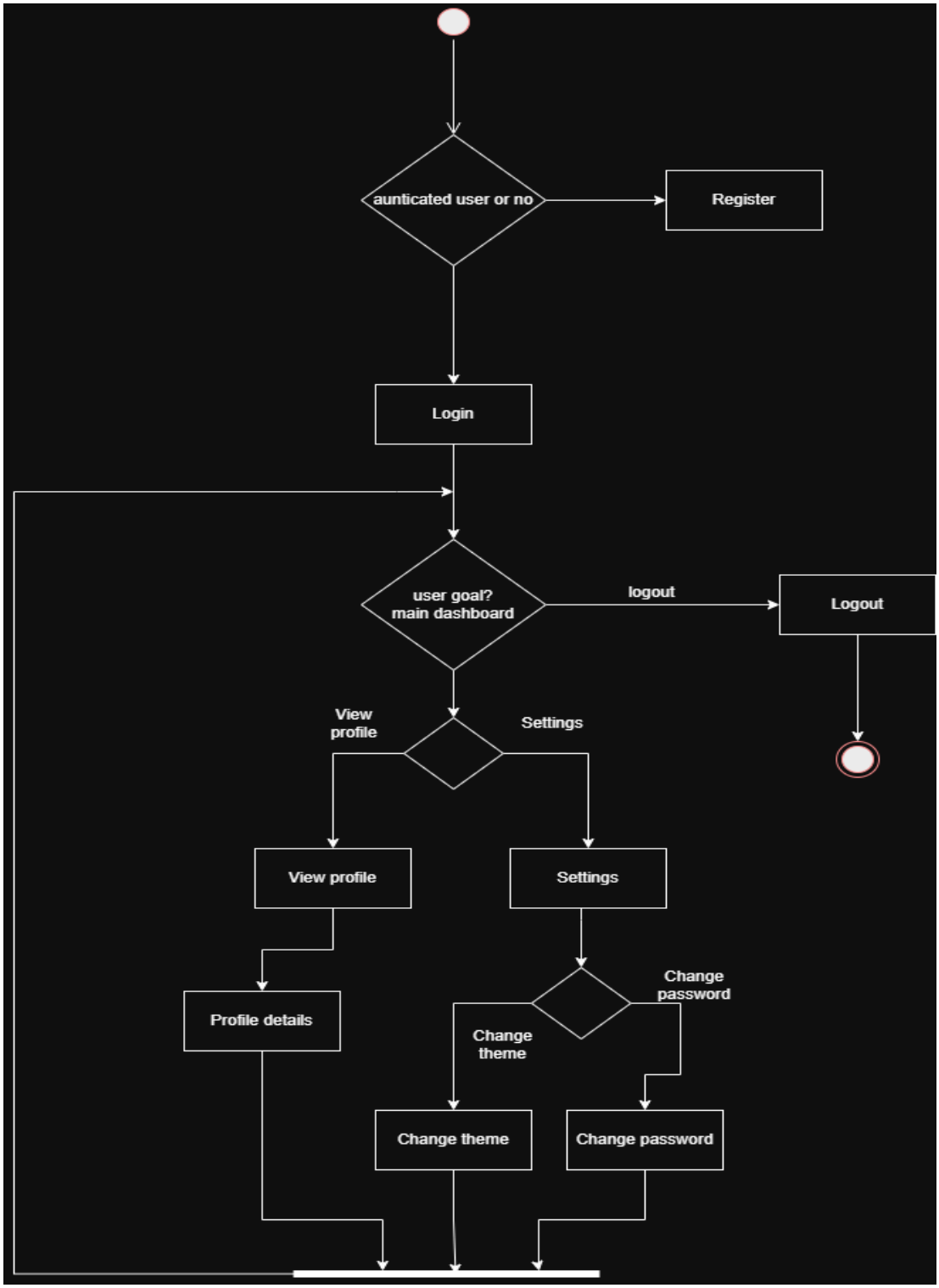
5.1. ERD



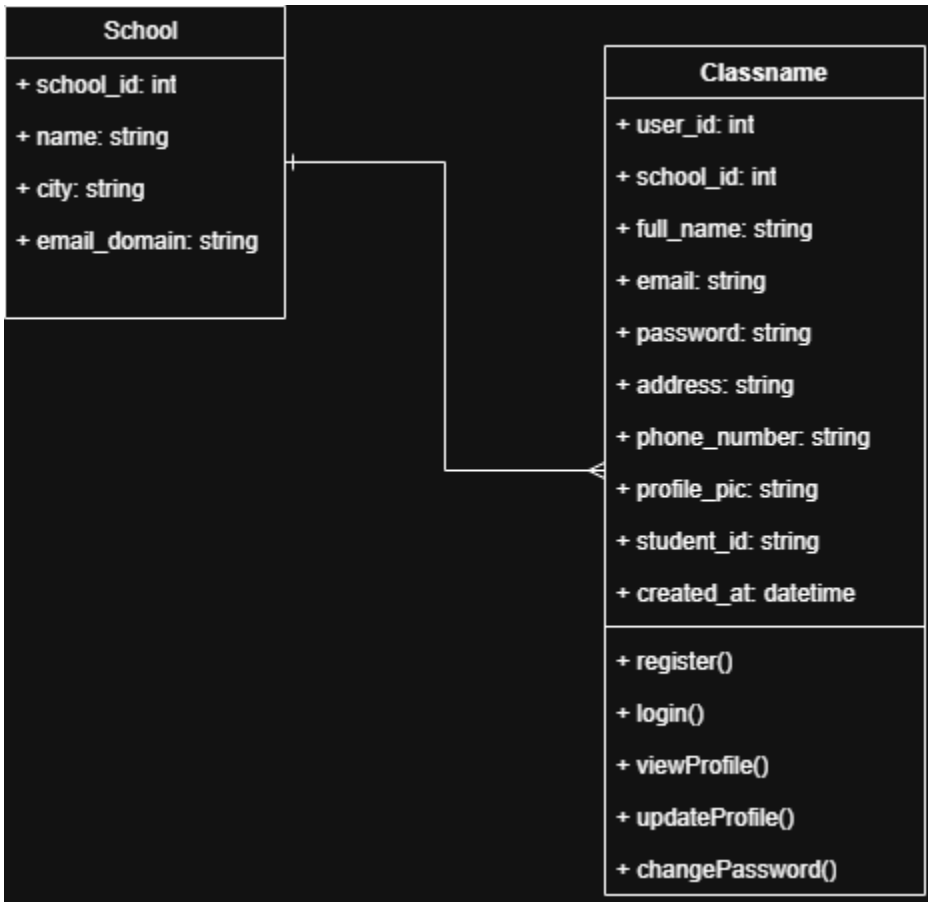
5.2. Use Case Diagram



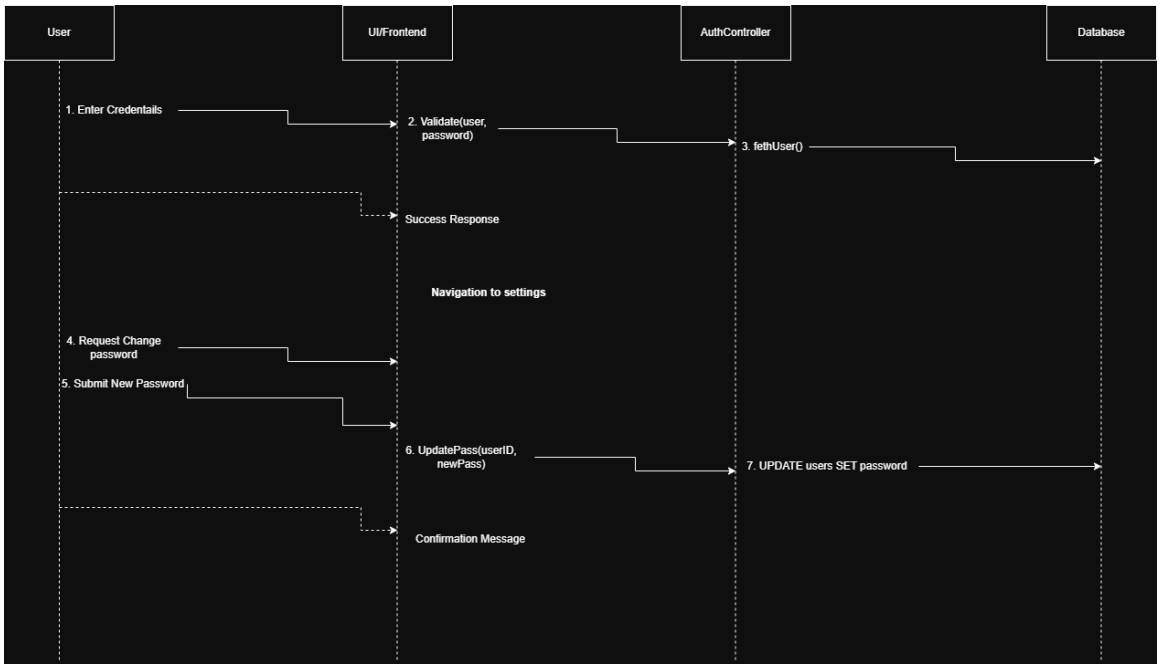
5.3. Activity Diagram



5.4. Class Diagram



5.5. Sequence Diagram



6. Appendices

- **Glossary: PK/FK** (Database keys), **JWT** (Secure tokens), **Bcrypt** (Encryption).
- **Tech Stack:** React/React Native, Spring Boot, Kotlin(android) and MySQL/PostgreSQL.
- **Diagrams:** Refer to attached **ERD**, **Use Case**, **Activity**, **Class**, and **Sequence** diagrams for technical logic.